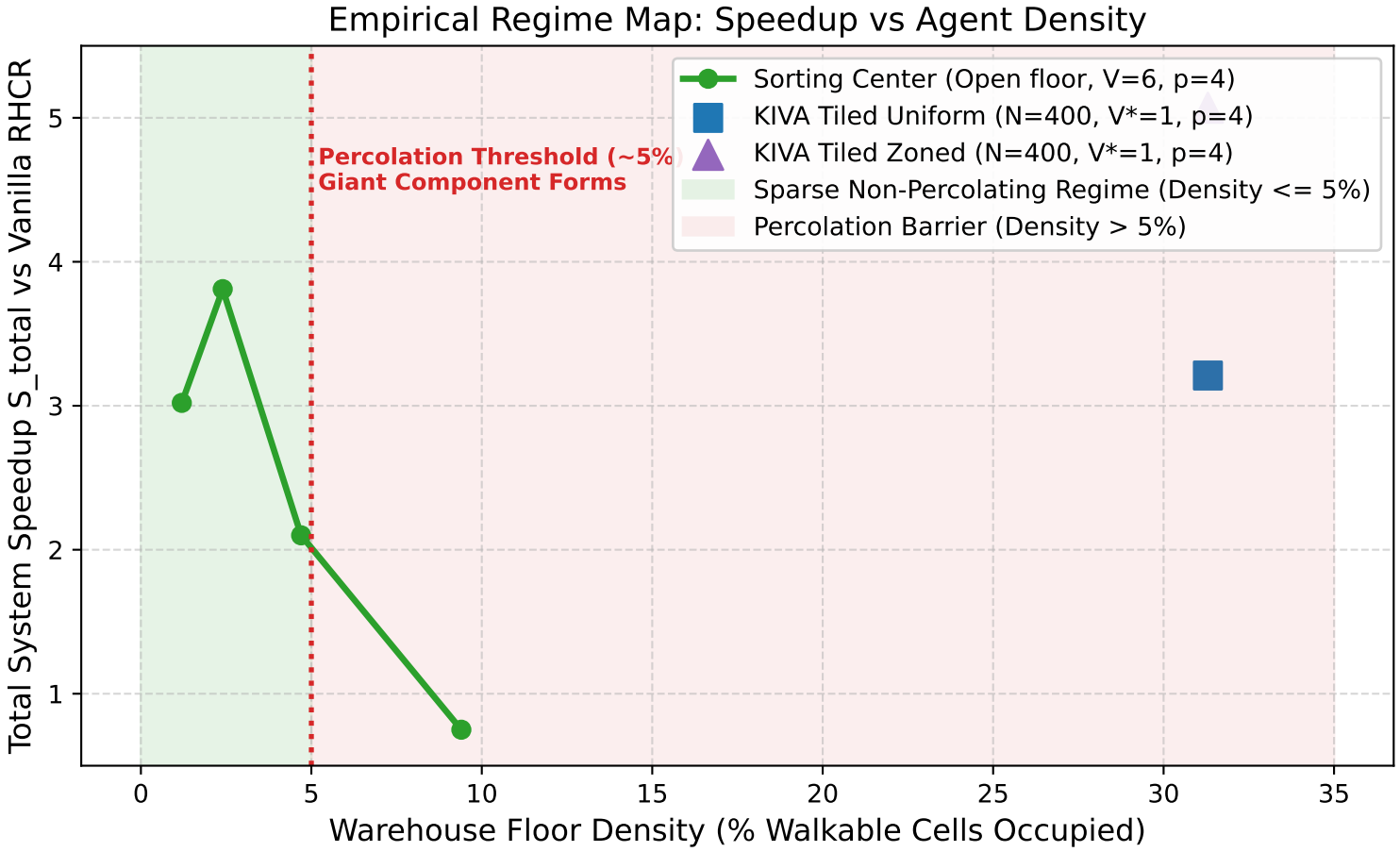
Empirical Regime Map: Speedup vs Agent Density
Total System Speedup S_total vs Vanilla RHCR
Warehouse Floor Density (% Walkable Cells Occupied)
Percolation Threshold (~5%)
Giant Component Forms
Sorting Center (Open floor, V=6, p=4)
KIVA Tiled Uniform (N=400, V*=1, p=4)
KIVA Tiled Zoned (N=400, V*=1, p=4)
Sparse Non-Percolating Regime (Density <= 5%)
Percolation Barrier (Density > 5%)
0
5
10
15
20
25
30
35
1
2
3
4
5